Lab #5: Hardware Implementation of a 4-Bit Calculator

In this lab, you will build the hardware of a 4-bit adder-subtractor using the logic gates AND, OR, NOT, and XORs. **This lab is to be done _individually_.**

Part 1: Optimizing Gate Usage

For Lab 4, you got the minimum SOP expressions for S_n (sum) and C_{n+1} (carry out) that define a full adder. However, since you are going to implement these equations in hardware later in the lab, it is in your best interest that you optimize both expressions as much as possible. In particular, you should minimize the number of *gates* required to implement the circuit. You may use all gates in your kit except the full adder chip. Below are the minimum SOP expressions for S_n and C_{n+1}.

$$S_n = A_n'B_n'C_n + A_n'B_nC_n' + A_nB_n'C_n' + A_nB_nC_n$$

$$C_{n+1} = A_nB_n + A_nC_n + B_nC_n$$

Questions:

1. Count how many **2-input** gates are required to implement the minimum SOP of S_n (above), including 3 NOT gates.

2. Count how many 2-input gates are required to implement the minimum SOP of C_{n+1} (above).

3. If you built both circuits, how many gates would you need for *one* full adder? What about *four* full adders?

4. Prove that $S_n = A_n \oplus B_n \oplus C_n$. Show your work. How many 2-input gates are required to implement this?

5. Prove that $C_{n+1} = A_nB_n + C_n(A_n \oplus B_n)$. Show your work. (Hint: try expanding $A_n + B_n \rightarrow A_n'B_n + A_nB_n' + A_nB_n$). How many 2-input gates are required to implement this?

6. Can any gates be shared between the new expressions for S_n and C_{n+1}? If they share, how many 2-input gates are required to implement one full adder using these circuits?

7. Is the minimum SOP always the best expression to build in hardware?

Part 2: Simulating the Full-Adder Subtractor

Now, you will simulate a **4-bit** full adder-subtractor using your simplified design from Part 1. As in Part 3 of Lab 4, you will add XOR gates to complement B to allow the calculator to perform two's-complement subtraction when C_{in} is set to 1. You should use the following formulas for each full adder-subtractor:

$$S_n = A_n \oplus X_n \oplus C_n$$

$$C_{n+1} = A_n X_n + C_n(A_n \oplus X_n)$$

where $X_n = B_n \oplus C_{in}$ and $n \in \{0,1,2,3\}$. C_{in} is connected to input C_0, and C_4 is ignored. Note that $n = 0$ corresponds with the adder that produces the least significant bit (bit 0), and $n = 3$ corresponds with the adder that produces the most significant bit (bit 3).

You may use any digital logic simulator you like, but ideally one that allows for manual manipulation of the inputs.

Questions:

8. Build a simulated circuit with a 4-bit full adder-subtractor using the equations above (do not use an adder macro). Remember to reuse gate values between the two circuits in each adder! Make sure you add probes and/or labels to each A_n, B_n, and S_n, as well as C_{in}. Add a screenshot of your build.

9. What range of two's-complement values can be used in your 4-bit calculator?

10. Give four examples of $A + B$ using your calculator. (Hint: have $A > 0$ plus $B > 0$, $A > 0$ plus $B < 0$, $A < 0$ plus $B > 0$, and $A < 0$ plus $B < 0$). Place the inputs and results in a table and include *one* screenshot from your addition calculations.

11. Give four examples of $A - B$ using your calculator. (Hint: have $A > 0$ minus $B > 0$, $A > 0$ minus $B < 0$, $A < 0$ minus $B > 0$, and $A < 0$ minus $B < 0$). Make sure you place the inputs and results in a table and include *one* screenshot from your subtraction calculations.

Part 3: Building your SimUAid Design in Hardware

For the last part of the lab, you will build your simulated circuit design on your breadboard. Make sure that your circuit can perform addition and subtraction using two's complement binary arithmetic. For your hardware build, you are allowed to use any of your TTL chips except the Full-Adder chip. In your presentation, perform addition and subtraction operations (3 of each) selected from your examples in Question 10 and Question 11, respectively. Use LEDs to visualize the result. Note in the video which input switches and LEDs are the MSBs and the LSBs!

Questions:

12. What pull-up/pull-down resistor values are you using and why? If you are unsure, please see Lab 1, Part 2 and the offline TTL lecture in the Labs module on Canvas.

Demo notes:

Make sure that you put your name somewhere visible on your breadboard. Make sure you state your name in the video.

It is required that your video is continuous, i.e. there should be no pauses, edits or interruptions to the recording. However, you may speed it up as appropriate.

The demo must show you DISASSEMBLING your circuit in your video at the end of the demo. Failure to do so will result in a loss of 50% of your grade.

The demo must be no more than a 6-minute video. Add it to your Google Drive or YouTube and put a *publicly accessible* link to that video in your report.

Part 4: Summary of Deliverables

Turn in your report in either PDF or Word format, and make sure it is named:
ECE3441_ StudentID#_Lab5.

****Note: Please refrain from using notebook paper or writing in the attached lab. Make a separate Word document and have it submitted as a word or as a pdf document in Canvas. Points will be deducted if you do not follow instructions.**

Use the following structure for your lab report:

Course Name and Number – Semester

Lab Number and Title

Name

Overview: Goals of the Lab

Briefly (~ 1 small paragraph) outline the goal of this lab, as well as the work done.

Step-by-Step Procedures and Results

This must be in chronological order of how Lab 5 should be completed. Make sure you include all the necessary information that is mentioned in the lab and answer all questions.

Conclusion

Summarize what you have learned in this lab.